Structural Fatigue and Metastable Closure in Runtime Monitoring

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Abstract. Operational closure does not imply structural safety: admissibility degradation accumulates historically and is not reset on reclosure. This condition is metastable closure. STAK-PSAL detects it via $\mathcal{F}_n$, a cumulative fatigue metric invisible to instantaneous monitors. A formal layer (machine-checked Coq proofs of admissibility invariants, the contraction bridge, and the structural separation between operational closure and metastable closure) grounds an operational ε-admissible OCaml monitor. Metastably closed runs exhibit 13.8–14.9× higher $\mathcal{F}_n$ than stably recovered runs with identical runtime status; in the evaluated model family, the CRITICAL forecast yields zero false negatives across 2,159 ruptured runs with mean lead time 11.6 to 13.7 steps.

1 Runtime Preservation of Safety-Relevant Distinguishability

A runtime system may remain operationally closed while already possessing a bounded rupture horizon. A runtime monitor may certify closure even when the system has accumulated admissibility fatigue that renders future collapse substantially more likely. This danger is not visible in instantaneous outputs or anomaly scores: two systems may present identical runtime observations while differing radically in structural fatigue and future rupture trajectory.

We introduce STAK-PSAL (Structural Tension and Admissibility with a Parallel Structural Assurance Layer), a runtime monitoring architecture for detecting metastable closure under cumulative admissibility fatigue.

Table 1. Identical runtime snapshots; divergent structural states (empirical means, 7,800 runs)

Property	**Stable recovery (R3)**	**Metastable closure (R2)**
Current status	CLOSED	CLOSED
Instantaneous d_t	0.43	0.44
Output anomaly	nominal	nominal
Prior META_REVIEW episodes	0	≥1
Mean $\mathcal{F}_n$	0.165	2.462
Future rupture rate	0%	100%

Table 1 is not hypothetical: these are empirically observed properties across 7,800 evaluated runs; a monitor classifying by current status or instantaneous

d_t cannot distinguish them, but $\mathcal{F}_n$ can (Section 7). Current admissibility is instantaneous: it classifies the present reduction. Structural fatigue is historical: it records the cumulative cost required to preserve admissibility across prior destabilisations.

A cyber-physical system reduces state $\mathcal{X} = \mathbb{R}^n$ to observations $\mathcal{O} = \mathbb{R}^m$ via M_t; safety requires M_t to preserve all safety-relevant distinctions. Static verification certifies M_0 and says nothing about M_t for $t > 0$. Degradation collapses safety-distinct states into identical observations while every individual output remains in-distribution. Anomaly detectors miss this; output monitors detect deviations in $M_t(x)$ but not in $\mathrm{dist}(M_t(x), M_t(y))$.

Example 1 (Industrial pressure monitor). A 64-dimensional industrial sensor vector spans pressure, temperature, flow, and vibration modalities. The safety function $\Phi(z) = \text{ESTOP}$ iff $z_0 > 0.70$. Two systems operate under identical parameters ($\delta = 0.020$, $b = 0.10$): one achieves stable recovery (R3) and one re-enters the warning band after apparent reclosure, subsequently rupturing (R2). Both inputs remain in-distribution as W_t degrades (Figure 1). At the moment of reclosure both systems present identical runtime status; only $\mathcal{F}_n$ distinguishes them.

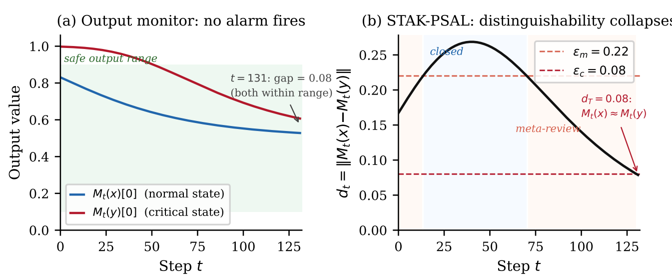

Fig. 1. Statistical normality may coexist with admissibility collapse. Output monitoring remains silent while d_t approaches ε_c.

The principal contribution is the identification, formal characterisation, and operational detection of *metastable closure*: a run classified as operationally closed while carrying a bounded rupture horizon. *First*, structural divergence under operational equivalence is machine-checked in the metastable recovery semantics (Section 8). *Second*, a bridge from strict to ε-admissibility: under geometric contraction, threshold crossing from ε to any $\varepsilon' < \varepsilon$ is guaranteed in bounded steps (Section 4.4). *Third*, $\mathcal{F}_n$ is validated as a predictive estimator: in the evaluated model family, CRITICAL has zero false negatives and mean lead time 11.6 to 13.7 steps across 7,800 runs (Sections 4–7). The monitor is scoped to continuous degradation; applicability beyond this regime is an open boundary (Section 11). Table 2 situates the contribution.

Table 2. Existing monitors and their blind spot for distinguishability collapse

Approach	What it detects	What it misses
Anomaly detection	unusual input traces	distinguishability collapse
Drift detection	distributional shift	safety-equivalence collapse
Uncertainty monitor	confidence loss	admissibility margin collapse
STAK-PSAL	distinguishability degradation	out-of-scope degradation

2 Failure of Static Admissibility

2.1 Deployment Architecture Assumptions

STAK-PSAL presumes white-box access to M_t. It also presumes an observational enrichment pathway: either mutable access to the weight matrix or an equivalent runtime resource such as secondary sensing, adaptive sampling, or operator escalation.

2.2 The Degradation Trajectory

Under the weight update $W_{t+1} = (1-\delta)\,W_t$, the separation d_t decreases monotonically toward 0 (Figure 1b). At $\delta = 0.020$, rupture occurs at step 114; the horizon is a function of degradation rate and initial margin, neither readable from the initial admissibility certificate. Verification of M_0 does not certify M_{71}.

2.3 The Enrichment Response as Observational Resolution Enhancement

Enrichment is the operational escalation triggered on entry to the warning band. In the demonstrator it is realised as a targeted weight boost: on detecting that d_t has fallen into $(\varepsilon_{\mathrm{c}}, \varepsilon_{\mathrm{m}}]$, the dimension $j^* = \arg\max_j |x_j - y_j|$ in row 0 of W_t receives a one-time additive boost $b > 0$.

The weight-boost is one instance of a broader family of observational resolution enhancements; Table 3 lists mechanisms by deployment class. The theoretical content is independent of the particular mechanism.

Table 3. Enrichment mechanisms by deployment class

Deployment class	Mechanism	Realisation
Trusted / upgradeable	Matrix modification	additive boost in row 0 (demonstrator)
	Adaptive sampling	raise polling rate on discriminating dim.
Adversarial / certified	Sensor escalation	secondary sensor engagement
	Modality augmentation	corroborating modality activation
	Operator escalation	operator notification

For adversarial or certified deployments, the immutable variants in Table 3 are not optional engineering alternatives but the required deployment models: mutable matrix modification is appropriate only in trusted demonstrators or authenticated upgradeable environments. The formal admissibility obligations are unchanged under these substitutions, although each deployment mechanism requires its own implementation proof.

Enrichment restores ε-admissibility if d_t rises above ε_{m}; sufficient boost ($b \geq b_{\mathrm{crit}}(\delta)$) sustains it indefinitely, insufficient boost delays but does not prevent rupture.

Enrichment budget. Enrichment fires once per META_REVIEW entry; the boosted weight erodes at rate δ, injecting a margin that subsequently decays. The operational layer therefore has finite enrichment capacity: repeated escalation consumes observational elasticity and may itself induce metastable behaviour. A reclosed system will reach the warning threshold again sooner than one never destabilised. This budget asymmetry is the operational mechanism behind the MRP (Section 8); adversarial implications are in Section 9.

3 Admissibility Definitions

Let $\mathcal{X} = \mathbb{R}^n$, $\mathcal{O} = \mathbb{R}^m$, $m \ll n$. Let $\mathcal{A}$ be a finite action set and $\Phi : \mathcal{X} \to \mathcal{A}$ a safety function.

Definition 1 (Reduction operator). *$M_t : \mathcal{X} \to \mathcal{O}$; in the demonstrator $M_t(z) = \sigma(W_t \cdot z)$ for $W_t \in \mathbb{R}^{2\times 64}$.*

Definition 2 (Admissibility (strict)). *M_t is admissible w.r.t. Φ if $M_t(x) = M_t(y) \Rightarrow \Phi(x) = \Phi(y)$.*

Definition 3 (ε-admissibility). *M_t is ε-admissible if*

$$\Phi(x) \neq \Phi(y) \;\Rightarrow\; \mathrm{dist}(M_t(x), M_t(y)) > \varepsilon.$$

Definition 2 is the $\varepsilon \to 0$ limit; the formal layer reasons over Definition 2, the operational layer uses Definition 3 with $\varepsilon = \varepsilon_{\mathrm{c}}$. The bridge is in Section 4.4.

Definition 4 (Boundary, rupture, closure). *A* boundary *is $B \subseteq \mathcal{X}$ with $x \in B, y \notin B \Rightarrow \Phi(x) \neq \Phi(y)$; M_t is* boundary-preserving *if $x \in B, y \notin B \Rightarrow \mathrm{dist}(M_t(x), M_t(y)) > \varepsilon_{\mathrm{c}}$ (in Example 1, $B = \{z \mid z_0 > 0.70\}$). A* rupture *at t^* is a pair (x, y) with $\Phi(x) \neq \Phi(y)$ and $\mathrm{dist}(M_{t^*}(x), M_{t^*}(y)) \leq \varepsilon_{\mathrm{c}}$; no correct safety decision is recoverable from $M_{t^*}(x)$ alone. The system is* closed *at t if M_t is ε-admissible and $\mathrm{dist}(M_{t+1}, M_t) \leq \eta$ for $\eta > 0$ (current admissibility plus local stability).*

Definition 5 (Rupture certificate). *A* rupture certificate *is a triple (x, y, t^*) such that (x, y) is a rupture at t^*, with the following three witnesses. The* logical witness *establishes $\mathrm{dist}(M_{t^*}(x), M_{t^*}(y)) \leq \varepsilon_{\mathrm{c}}$ and $\Phi(x) \neq \Phi(y)$. The* operational witness *records the divergent actions with their physical interpretations. The* instability witness *records the sequence of boundary revisions, oscillation count, and peak tension preceding t^*.*

Definition 6 (Runtime estimator). *A* runtime estimator *is a computable function $\mathcal{E}$: States $\times$ History $\to \mathbb{R}$ that tracks a structural property; it signals proximity to rupture without certifying it.*

Table 4 gives the RV reading of each term.

Table 4. STAK-PSAL terminology in standard RV reading

STAK-PSAL term	RV reading
ε-admissibility	runtime-checkable separation invariant
Rupture	monitor indistinguishability failure
Metastable closure	confirmed recovery with accumulated admissibility deficit
CRITICAL forecast	anticipatory violation signal; leading indicator

4 The Three-State Monitor

Monitor contract. The monitor inputs are (d_t, $status_{t-1}$, W_t); the output is ($status_t$, $cert$), with $cert$ populated iff $status_t =$ HARD_RUPTURE. The invariant is: CLOSED status implies M_t is ε-admissible (Theorem 1, (T4)). Escalation from CLOSED occurs on $d_t \in (\varepsilon_c, \varepsilon_m]$; escalation from META_REVIEW occurs on $d_t \le \varepsilon_c$. The CRITICAL signal precedes every confirmed HARD_RUPTURE (zero false negatives, 7,800 runs) with mean lead time 11.6 to 13.7 steps.

The monitor maintains a weight matrix $W_t \in \mathbb{R}^{2\times 64}$ and a fixed critical pair (x, y) with $\Phi(x) \ne \Phi(y)$. The reduction is $M_t(z) = \sigma(W_t \cdot z)$. On each step, weights decay: $W_{t+1} = (1-\delta)\, W_t$. On CLOSED→META_REVIEW transition, enrichment fires once: the dimension $j^* = \arg\max_j |x_j - y_j|$ in row 0 of W_t receives a one-time boost $b > 0$.

CLOSED ($d_t > \varepsilon_m$): ε-admissible, no action. META_REVIEW ($\varepsilon_c < d_t \le \varepsilon_m$): admissible but degrading; enrichment fires once on entry, autonomous execution restricted; reclosure if $b \ge b_{\text{crit}}$, otherwise d_t falls to rupture. HARD_RUPTURE ($d_t \le \varepsilon_c$): admissibility failed; certificate emitted (Definition 5), monitoring halts. Thresholds: $\varepsilon_c = 0.08$, $\varepsilon_m = 0.22$ ($d_0 \approx 0.72$).

4.1 State Transition Semantics

Table 5 gives the complete state machine.

Table 5. State transitions (C = CLOSED, M = META_REVIEW, H = HARD_RUPTURE)

From	Condition on d_t	To	Action
C	$d_t > \varepsilon_m$	C	none
C	$\varepsilon_c < d_t \le \varepsilon_m$	M	Enrichment fires on j^*
C	$d_t \le \varepsilon_c$	H	Certificate emitted; halt
M	$d_t > \varepsilon_m$	C	Reclosure recorded
M	$\varepsilon_c < d_t \le \varepsilon_m$	M	Estimators updated; no enrichment
M	$d_t \le \varepsilon_c$	H	Certificate emitted; halt
H	all	H	Terminal state

4.2 The Enrichment/Degradation Balance and b_{crit}

The critical boost value $b_{\text{crit}}(\delta)$ is the minimal b for which stable-recovery outcomes (reclosure without subsequent rupture) constitute a majority at 100 seeds. Experimentally, b_{crit} scales linearly:

$$b_{\text{crit}}(\delta) \approx 10\delta \tag{1}$$

Valid across $\delta \in \{0.010, \ldots, 0.030\}$; not extrapolated beyond the evaluated regime.

4.3 Formal Invariants and Their Operational Role

The following invariants are machine-checked over Definition 2. They constrain the structural relations between admissibility, boundary preservation, closure, and the rupture certificate; the operational monitor instantiates these as ε-regime approximations.

Theorem 1 (Coq invariants). *Over Definitions 2–5 the following hold:*

$$\textit{(T1)}\ \texttt{admissible\_not\_ruptured} : \mathit{Admissible} \Rightarrow \neg\, \mathit{Rupture}$$

$$\textit{(T2)}\ \texttt{boundary\_preservation} : \mathit{BoundaryPreserved} \Rightarrow \mathit{Admissible}$$

$$\textit{(T3)}\ \texttt{certificate\_implies\_rupture} : \mathit{RuptureCertificate} \Rightarrow \mathit{Rupture}$$

$$\textit{(T4)}\ \texttt{closure\_excludes\_rupture} : \mathit{Closure} \Rightarrow \neg\, \exists\, \mathit{RuptureCertificate}$$

$$\textit{(T5)}\ \texttt{rupture\_implies\_violation} : \mathit{Rupture} \Rightarrow \neg\, \mathit{BoundaryPreserved}$$

A direct corollary of (T3) is RuptureCertificate $\Rightarrow \neg$ Admissible.

4.4 Operational Approximation and ε-Admissibility

The formal theorems and operational monitor occupy different mathematical regimes (Table 6): theorems over strict admissibility (Definition 2), the monitor over ε-admissibility (Definition 3).

Table 6. Two layers of the monitor architecture

Formal layer	**Operational layer**
Strict equality $M(x) = M(y)$	ε margin $\mathrm{dist}(M(x), M(y)) > \varepsilon$
Theorem space	Continuous runtime monitoring
Exact admissibility	Approximate admissibility
Coq proofs	Empirical OCaml prototype
Boundary conditions for the monitor	Step-by-step runtime decisions

The machine-checked semantic hierarchy fixes structural relations at the strict-equality limit; the OCaml monitor approximates these at finite ε. The machine-checked existential bridge (`formal/threshold_bridge.v`) connects them: under geometric contraction, if $d_t \le \varepsilon$, then for any $\varepsilon' \in (0, \varepsilon)$ there exists k such that $d_{t+k} \le \varepsilon'$. The explicit logarithmic ceiling bound on the horizon T stated below is a quantitative refinement of that machine-checked existential.

Lemma 1 (Operational rupture bound). *Let $M_t(z) = \sigma(W_t z)$ with $W_{t+1} = (1-\delta)W_t$ and σ Lipschitz with constant L_σ. If $d_t = \mathrm{dist}(M_t(x), M_t(y)) \le \varepsilon$, then for any $\varepsilon' \in (0, \varepsilon)$ the horizon $T = t' - t$ at which $d_{t'} \le \varepsilon'$ satisfies*

$$T \;\le\; \left\lceil \frac{\log(\varepsilon/\varepsilon')}{\log(1/(1-\delta))} \right\rceil \;\approx\; \left\lceil \frac{1}{\delta}\, \log \frac{\varepsilon}{\varepsilon'} \right\rceil .$$

The operational regime at threshold ε collapses into the regime at any $\varepsilon' < \varepsilon$ in bounded, explicitly computable time; the strict regime $d_{t'} \to 0$ is the asymptotic limit.

Proof (Sketch). Multiplicative decay of W_t gives $|W_t(x-y)| \propto (1-\delta)^{t'-t}$. Lipschitz σ yields $d_{t'} \le (1-\delta)^{t'-t} d_t / c$ for c absorbing the Lipschitz factor and projection geometry; solving $(1-\delta)^T \varepsilon \le \varepsilon'$ gives the bound.

The machine-checked existential bridge guarantees threshold crossing in bounded steps; the explicit logarithmic ceiling bound is the quantitative refinement isolating the precise horizon.

4.5 Bounding the Operational Approximation

As δ increases, the predictive horizon shrinks: $T \approx (1/\delta)\log(\varepsilon/\varepsilon')$ (Lemma 1). The stochastic sweep (Section 7.3) validates this bound empirically. At $\delta = 0.015$, mean CRITICAL lead time is 16.8 steps; at $\delta = 0.030$ it collapses to 9.4 steps, with a minimum of 1 step in the most rapid degradation cases. This matches the asymptotic expectation: at $\delta = 0.030$ the bound $T = (1/0.030)\log(\varepsilon_c/\varepsilon') \approx 33$ steps to strict collapse is consistent with the observed compressed lead-time distribution.

5 Predictive Instability Estimation

The four signals below are runtime estimators (Definition 6), each active several steps before d_t crosses ε_c. Of these, $\mathcal{F}_n$ is architecturally primary: the only signal that accumulates across episodes rather than evaluating instantaneous state. Instantaneous admissibility cannot distinguish metastable closure from stable recovery; $\mathcal{F}_n$ operationalises cumulative structural degradation invisible to state-local monitoring.

Tension slope. Let T_t be the tension score (weighted EMA of the separation deficit).

$$\dot{T}_t = \alpha_s(T_t - T_{t-1}) + (1-\alpha_s)\dot{T}_{t-1}, \quad \alpha_s = 0.25 \tag{2}$$

Separation EMA.

$$\bar{d}_t = \alpha_d\, d_t + (1-\alpha_d)\,\bar{d}_{t-1}, \quad d_t = \mathrm{dist}(M_t(x), M_t(y)), \quad \alpha_d = 0.10 \tag{3}$$

Rupture pressure.

$$R_t = \frac{1}{d_t + \varepsilon_0}, \quad \varepsilon_0 = 0.001 \tag{4}$$

Recovery elasticity. Let T_{entry} be the tension at META_REVIEW entry and Δt the steps elapsed since enrichment fired.

$$E_r^{(t)} = \frac{T_{\text{entry}} - T_t}{\Delta t} \tag{5}$$

$E_r^{(t)} > 0$ indicates recovery in progress; $E_r^{(t)} < 0$ indicates degradation outpacing enrichment. Defined only for `steps_in_meta` > 2.

Definition 7 (Adversarial elasticity inversion). *An* adversarial elasticity inversion *occurs when $E_r^{(t)} < 0$ and $|\Delta P_t| \le \eta_P$ over a sustained* META_REVIEW *interval, where P_t denotes the monitored physical degradation proxy and η_P is* *the expected physical drift bound. In this case, loss of recovery elasticity is not explained by the monitored physical degradation proxy. The remaining explanation is exhaustion of the enrichment pathway, which may be benign or adversarial depending on provenance.*

The compound condition $E_r^{(t)} < 0$ with bounded $|\Delta P_t|$ and repeated META_REVIEW entry constitutes a bounded rupture claim of potentially adversarial origin (Section 9). Note that negative elasticity alone is necessary but not sufficient for rupture: 100% of R2 runs exhibit it, but so do 20% of R3 runs (Table 9). The adversarial interpretation requires the additional condition of bounded physical drift.

Cumulative fatigue.

$$\mathcal{F}_{t+1} = \mathcal{F}_t + \lambda_1 |\Delta T_t| + \lambda_2 \max(0, -E_r^{(t)}) + \lambda_3 \max(0, R_t - R_0) \tag{6}$$

Accumulated during META_REVIEW steps only ($\lambda_1 = 1.0$, $\lambda_2 = 0.1$, $\lambda_3 = 0.01$, $R_0 = 5.0$); not reset on reclosure. $\mathcal{F}_n$ is not an anomaly magnitude. It is a memory variable that records the cumulative enrichment expenditure required to preserve runtime admissibility.

Definition 8 (CRITICAL forecast). *The system is in* CRITICAL *state at t when:*

$$\dot{T}_t > \theta_s \ \wedge \ R_t > R_c \ \wedge \ E_r^{(t)} < E_w \ \wedge \ \texttt{steps\_in\_meta} > 2 \tag{7}$$

where $\theta_s = 0.002$, $R_c = 10.0$, $E_w = 0.005$ (calibrated on the 13-run deterministic sweep; stochastic and benchmark results are out-of-calibration evaluations). CRITICAL *signals closing intervention window; a system satisfying* (7) *may still recover at higher boost levels.*

6 OCaml/Coq Implementation

The monitor is implemented in OCaml 4.14; formal invariants are machine-checked in Coq 8.18. Table 7 lists module responsibilities.

Table 7. Module responsibilities

Module	Responsibility
`types.ml`	Core types and status variants
`lnp.ml`	Reduction $M_t = \sigma(W_t \cdot z)$; degradation; enrichment
`admissibility.ml`	Φ; three-way check (`Admissible` / `Drifting` / `HardRupture`)
`tension.ml`	Estimators: $\bar{d}_t$, R_t, $\dot{T}_t$
`machine.ml`	State transitions; $E_r^{(t)}$; $\mathcal{F}_n$
`instability.ml`	STABLE/STRAINED/CRITICAL classification
`certificate.ml`	Rupture certificate builder and JSON serialiser
`scenario.ml`	Four-modality industrial benchmark scenario
`formal/stak_psal.v`	Coq proofs of invariants (T1)–(T5) of Theorem 1
`formal/contraction_core.v`	Coq proof of geometric multi-step contraction lemma
`formal/threshold_bridge.v`	Coq proof of finite ε-to-ε' threshold crossing (Lemma 1, existential form)
`formal/metastable_recovery.v`	Coq proofs of structural separation, admissible visibility failure, and certificate soundness hierarchy

Each step computes $(d_t, \bar{d}_t, \dot{T}_t, R_t, E_r^{(t)})$ followed by state transition and optional certificate emission. Enrichment fires once per CLOSED→META_REVIEW transition; the certificate builder populates all three witnesses of Definition 5 and serialises to JSON.

7 Experimental Evaluation

The experiments establish two claims: (1) metastably closed runs are distinguishable from stably recovered runs by $\mathcal{F}_n$ alone; (2) CRITICAL provides advance warning before confirmed collapse. All three settings share the same parameters: $\varepsilon_c = 0.08$, $\varepsilon_m = 0.22$, $W_0[0][0] = 5.0$, $n = 64$, $m = 2$.

7.1 Representative trace

A representative R2 run ($\delta = 0.020$, $b = 0.10$, seed 42) demonstrates the principle: the two CLOSED intervals are indistinguishable under instantaneous d_t but separable by cumulative fatigue $\mathcal{F}_n$ (2.94 at rupture, 17.8× the R3 mean); CRITICAL fires 12 steps before. Representative metastable trajectories exhibit: (i) temporary reclosure, (ii) renewed decline in d_t, (iii) negative elasticity, and (iv) cumulative fatigue accumulation despite apparent recovery.

Figure 2 shows a paired R3/R2 run: both reclose at $t = 61$ with identical d_t and status; R2 re-enters META_REVIEW and ruptures at $t = 121$, CRITICAL firing 11 steps before, with structural divergence already visible in $\mathcal{F}_n$.

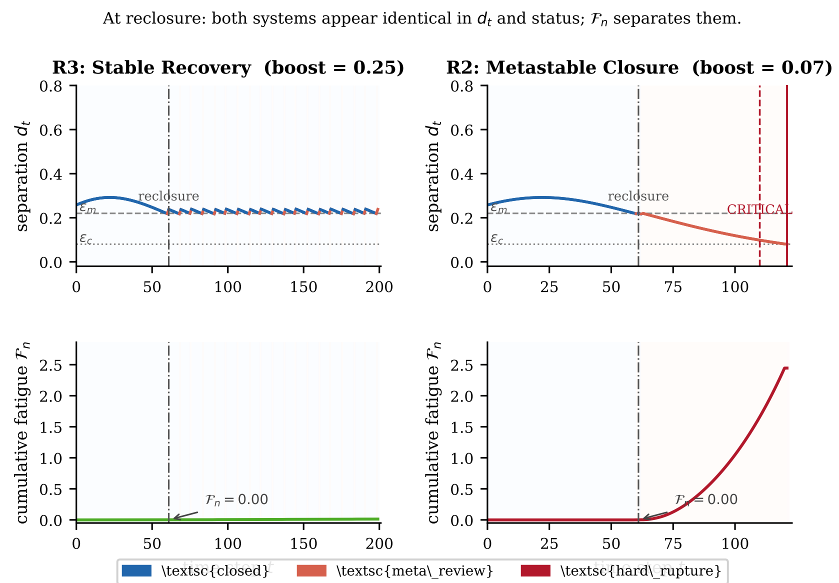

Fig. 2. R3 (stable recovery, left) and R2 (metastable closure, right) with the same seed and identical parameters except boost (0.25 vs. 0.05). Both reclose at $t = 61$; d_t and status are indistinguishable at that moment. $\mathcal{F}_n$ (bottom row) separates them: R2 carries substantially higher cumulative fatigue into its second episode.

7.2 Deterministic bifurcation sweep

Single noise-free input pair, $\delta = 0.020$, $b \in \{0.00, 0.05, \ldots, 0.60\}$ (13 runs), $T = 200$ steps. Three regimes appear at $b_{\text{crit}} = 0.20$: R1 ($b = 0$, rupture step 114), R2 ($b \in [0.05, 0.15]$, reclosed once then ruptured at steps 118 to 187), R3 ($b \geq 0.20$, stable). CRITICAL fires iff the run ruptures: 0 FP, 0 FN, mean lead time 11.5 steps (range 10 to 14) on 13 runs; confirmed at scale in Section 7.3.

7.3 Stochastic sweep

100 seeds per cell, 13 boost values, five degradation rates δ from 0.010 to 0.030 in steps of 0.005, $T = 200$ steps. Box-Muller noise on dims 1–63. Total: 6,500 runs.

Consistent with (1), each +0.005 step in δ requires +0.05 in b to sustain a majority of R3 outcomes: $b_{\text{crit}} \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$ for $\delta \in \{0.010, 0.015, 0.020, 0.025, 0.030\}$. Across 6,500 runs: 156 R0 (never entered META_REVIEW), 10 R1 (immediate failure), 1,071 R2 (metastable), 5,263 R3 (stable recovery).

Table 8. CRITICAL confusion matrices (stochastic and benchmark sweeps)

	Stochastic (6,500 runs)		**Benchmark (1,300 runs)**	
	Ruptured	Not ruptured	Ruptured	Not ruptured
CRITICAL fired	1,959 (TP)	150 (FP)	200 (TP)	0 (FP)
CRITICAL not fired	0 (FN)	4,391 (TN)	0 (FN)	1,100 (TN)

In the stochastic sweep recall is 100% and precision 92.9%; in the benchmark both are 100%. The 150 stochastic false positives are conservative interventions: at the moment CRITICAL fires, a recovering run is indistinguishable from one that will rupture. The cost of a false positive is unnecessary inspection; that of a false negative is an undetected rupture. Mean lead time is 11.6 steps overall (median 9; range 1 to 77), decreasing with δ from 16.8 steps at $\delta = 0.015$ to 9.4 at $\delta = 0.030$. The minimum of 1 step at $\delta \geq 0.025$ is the failure mode noted in Section 11: very rapid degradation can reach ε_c before (7) activates.

Table 9. Fatigue and elasticity by regime (stochastic sweep)

Regime	n	Mean $\mathcal{F}_n$	$\mathcal{F}_n$/R3	Pr(min $E_r < 0$)
R1: immediate failure	888	2.454	14.9×	100%
R2: fragile (metastable)	1,071	2.462	14.9×	100%
R3: stable recovery	4,385	0.165	1×	20%

Observation 2. *Negative elasticity is necessary but not sufficient for rupture: 100% of R2 and R1 runs exhibit* $\min E_r < 0$*, but 20% of R3 runs do as well.*

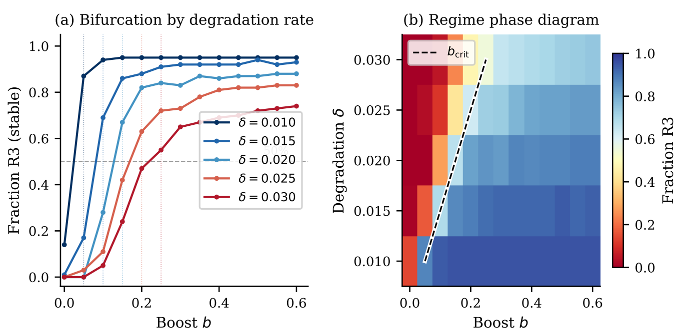

Fig. 3. Regime structure (6,500 stochastic runs). Left: R3 fraction vs. boost b for each δ; dotted lines mark $b_{\text{crit}}(\delta)$. Right: phase diagram; blue = R3, red = R1/R2. The $b_{\text{crit}} \approx 10\delta$ scaling appears as the diagonal boundary.

7.4 Multi-modal industrial benchmark

The 64D input is partitioned into four sensor modalities (16 dims each) matching Example 1. Anchor coordinates: $x_0 = 0.15, x_{16} = 0.20, x_{32} = 0.55, x_{48} = 0.10$ (normal); $y_0 = 0.88, y_{16} = 0.72, y_{32} = 0.22, y_{48} = 0.76$ (critical); $\Phi(z) =$ ESTOP iff $z_0 > 0.70$; sensor noise $\mathcal{N}(0, 0.05^2)$ on non-anchor dims. Per-modality rates: pressure $\delta_p = 0.020$, vibration $\delta_v = 0.015$, temperature $\delta_T = 0.012$, flow $\delta_f = 0.007$. Pressure dominates effective degradation; temperature and flow degrade more slowly, preserving corroborating boundary evidence longer. Sweep: 100 seeds × 13 boost values, $T = 200$ steps, 1,300 runs total.

Observation 3. $b_{\text{crit}} = 0.10$ *in the benchmark, lower than 0.15 at uniform* $\delta = 0.020$ *because slower degradation of non-pressure modalities extends the corroborating evidence window.*

Benchmark recall and precision are both 100% (right two columns of Table 8); mean lead time is 13.7 steps (range 7 to 24).

Table 10. Cross-setting comparison of key metrics

Metric	**Deterministic** (13 runs)	**Stochastic** (6,500 runs)	**Benchmark** (1,300 runs)
Total runs	13	6,500	1,300
Ruptured runs	8	1,959	200
Recall (CRITICAL)	100%	100%	100%
Precision (CRITICAL)	100%	92.9%	100%
Mean lead time (steps)	11.5	11.6	13.7
$\mathcal{F}_n$ ratio R2/R3	n/a	14.9×	13.8×
b_{crit}	0.20	0.15 (δ=0.020)	0.10

Observation 4. *Recall is 100% in all three settings; precision varies from 92.9% (stochastic, where noise permits* CRITICAL *in runs that recover) to 100% (benchmark, where multi-modal degradation yields cleaner separation).*

Reproducibility. All experiments were generated from the publicly available implementation at `https://github.com/dhwcmoore/Structural-Fatigue-and-Metastable-Closure-in-Runtime-Monitoring`; the repository includes the OCaml state machine, stochastic sweep, fatigue metrics, and plotting scripts required to reproduce all reported figures and lead-time statistics.

8 Metastable Closure and the Recovery Principle

Definition 9 (Metastable closure). *A run is* metastably closed *if it achieves at least one reclosure (*`rcls_t` $\neq -1$*), subsequently ruptures (*`rupt_t` $\neq -1$*), and has positive cumulative fatigue* $\mathcal{F}_n$(`rupt_t`) > 0*. Metastable closure corresponds to Regime 2 in the experimental classification.*

A metastably closed run is not a run that almost stabilised: it *did* stabilise, appeared closed, passed any instantaneous admissibility check, and then ruptured. R2 and R3 trajectories are separable only by $\mathcal{F}_n$ (Table 1).

Let $\mathcal{E}_k$ denote episode k (one META_REVIEW entry-to-exit cycle), with $d_{\min}^{(k)}$ the minimum separation EMA, $\mathcal{F}^{(k)}$ the fatigue accumulated, and $\bar{E}_r^{(k)}$ the mean recovery elasticity during $\mathcal{E}_k$.

The following principle is model-supported, not universally proved: the structural separation and certificate semantics are machine-checked; the quantitative claims (i)–(iii) are empirical over the evaluated STAK-PSAL model family.

Principle 5 (Metastable Recovery Principle). *Under the STAK-PSAL model with* $\delta > 0$ *and* $b < b_{\text{crit}}(\delta)$*, the following hold for metastably closed runs (Definition 9).*

Claim (i), persistent fatigue accumulation: $\mathbb{E}[\mathcal{F}_n \mid \text{R2}] \gg \mathbb{E}[\mathcal{F}_n \mid \text{R3}]$.

Claim (ii), elasticity depletion: $\min_t E_r^{(t)} < 0$ *for every metastably closed run.*

Claim (iii), episode-level trend (majority): *among metastably closed runs with $K \geq 2$ episodes, a strict majority exhibit both $d_{\min}^{(k+1)} \leq d_{\min}^{(k)}$ and $\mathcal{F}^{(k+1)} \geq \mathcal{F}^{(k)}$ for all k.*

Claim (iv), CRITICAL coverage: *no metastably closed run in any evaluated setting terminates in* HARD_RUPTURE *without having first satisfied* (7).

Proof (Empirical support). (i) follows from Table 9: R2 mean $\mathcal{F}_n = 2.462$, R3 mean $= 0.165$, ratio 14.9×. (ii) follows from the same table: $\min E_r < 0$ for 100% of R2 runs. (iii) follows from Table 11: majorities exhibit both decreasing $d_{\min}^{(k)}$ and non-decreasing $\mathcal{F}^{(k)}$ across episodes. (iv): in 2,159 ruptured runs, none reached HARD_RUPTURE without first satisfying (7).

Machine-checked metastable recovery semantics. Three results in `formal/metastable_recovery.v`. (i) `recovered_not_metastable`: *recovered*(t) and *metastable_closed*(t) are structurally disjoint. (ii) `admissible_visibility_failure`: there exists t at which the monitor reads CLOSED while the structural forecast is a bounded rupture claim. (iii) The certificate soundness hierarchy (`certificate_sound`, `certificate_within_horizon`, `certificate_marks_visible_metastability`): a constructive rupture witness is realisable within the declared horizon while the monitor reads CLOSED.

The empirical 13.8–14.9× fatigue ratio is not formally proved; it is an experimental observation over the evaluated model family.

Table 11. Episode-level monotonicity in R2 runs with $K \geq 2$ episodes ($n = 190$, multi-modal benchmark)

Condition	**Count**	**Fraction**
$d_{\min}^{(k+1)} < d_{\min}^{(k)}$ for all k	123/190	65%
$\mathcal{F}^{(k+1)} \geq \mathcal{F}^{(k)}$ for all k	118/190	62%
$\bar{E}_r^{(k+1)} < \bar{E}_r^{(k)}$ for all k	72/190	38%
All three simultaneously	72/190	38%

Separation and fatigue trend conditions hold for majorities; elasticity strict monotonicity (38%) is weaker since episode-to-episode noise disrupts strict ordering while the cumulative trend in claim (i) remains clean. A status-only monitor cannot distinguish a system that has never destabilised from one with $\mathcal{F}_n > 2$ (overwhelmingly R2); $\mathcal{F}_n \approx 0.165$ is an empirical discriminator, not a stability proof.

9 Security and Deployment Constraints

A runtime-modifiable reduction operator introduces an attack surface absent in immutable certified pipelines; three risk classes follow from the enrichment structure (Section 2). *Spoofing* of the warning condition: an adversary perturbing the input near the safety boundary may force the monitor into META_REVIEW and consume its enrichment budget, raising $\mathcal{F}_n$ without any physical degradation, consistent with the adversarial elasticity inversion identified in Section 5. *Adversarial activation* of the enrichment pathway: a compromised channel may inject distortions into row 0 of W_t that survive the immediate admissibility check but corrupt downstream behaviour. *Destabilisation by repeated probing*: a sequence of bounded perturbations near the boundary can deplete the enrichment budget, driving the system toward the R2 regime even when each individual perturbation is benign. The attack surface is therefore not the physical process alone, but the finite enrichment capacity of the observational boundary itself.

Adversarial exhaustion is more dangerous than direct drive to rupture: repeated bounded perturbations force META_REVIEW entry, consume enrichment capacity, and raise $\mathcal{F}_n$ while the physical process remains in bounds, producing adversarial elasticity inversion (Definition 7). Negative elasticity signals intrusion only under the compound condition of bounded $|\Delta P_t|$ and repeated META_REVIEW entry; 20% of R3 runs exhibit $\min E_r < 0$ in isolation (Table 9).

Mitigation is partial and deployment-dependent. Enrichment triggers should authenticate warning provenance; mutable pathways should be restricted to a rate-limited API surface; immutable variants (Table 3) preserve the monitoring contract without exposing the weight matrix. A full adversarial analysis is a defined open boundary of the present work.

10 Related Work

Runtime verification. Classical RV monitors execution traces against temporal specifications [1,2]. STAK-PSAL monitors the *reduction operator* M_t rather than the output trace, and anticipates violation before it is confirmed.

Drift detection. Conventional drift monitors evaluate deviation in current observation space. STAK-PSAL instead evaluates the cumulative structural cost required to maintain admissibility under repeated destabilisation.

Predictive and parametric monitoring. Parametric monitoring [4] and predictive monitoring [5] are the closest RV analogues; this paper instantiates predictive monitoring for admissibility degradation, with $\mathcal{F}_n$ as the domain-specific degradation functional.

Monitorability and observability. The monitorability question [15,16,17] asks what properties are monitorable and with what guarantee; STAK-PSAL specialises this to admissibility degradation, answering affirmatively for continuous degradation trajectories. Output-layer monitoring [7,3] assesses whether outputs are in-distribution; STAK-PSAL monitors M_t directly.

Cyber-physical systems, runtime assurance, and PHM. Runtime assurance for CPS [19,25] establishes runtime properties on autonomous systems with continuous dynamics; STAK-PSAL extends these patterns to the observation layer. Prognostics and Health Management encompasses degradation indicators [24] and latent fatigue accumulation [26]; $\mathcal{F}_n$ specialises the PHM fatigue functional to observational admissibility. The ecological distinction between bouncing back and structural recovery [11] maps directly onto the R2/R3 regime split.

Formal methods and static analysis. Static abstract interpretation [12] and CEGAR [13] establish soundness or refinement properties before deployment. STAK-PSAL monitors whether the runtime reduction continues to preserve the distinctions on which such properties depend.

11 Applicability, Limitations, and Domain Boundaries

Model scope and statistical claims. The current prototype uses a linear-nonlinear reduction with two critical vectors and scalar degradation; multiple concurrent critical pairs (x_i, y_i) are the immediate scaling challenge, requiring fatigue aggregation or per-pair tracking. Results are point estimates on a fixed model with calibrated thresholds and require confidence quantification. The CRITICAL thresholds were set on the 13 deterministic runs of Section 7.2; the stochastic and benchmark sweeps are out-of-calibration evaluations, not cross-validation. The zero-false-negative claim is model-specific: very rapid degradation can cause $d_t \leq \varepsilon_c$ before (7) activates (minimum lead time of 1 step at $\delta \geq 0.025$).

Proof and runtime gap. The Coq theorems use Definition 2 (exact equality); the OCaml monitor uses Definition 3. The threshold-crossing bridge is machine-checked in `formal/threshold_bridge.v` in its existential form: under geometric contraction, crossing from any ε to any smaller ε' is guaranteed in bounded steps. Closing the remaining gap formally requires restating the structural invariants (T1)–(T5) over Definition 3; the explicit logarithmic ceiling bound of Lemma 1 remains a quantitative refinement of the mechanised existential.

Domain boundary: continuous fatigue mathematics and discrete generative systems. STAK-PSAL rests on assumptions fitting continuous degradation: multiplicative weight decay, a smooth separation margin, an additive enrichment response, and a continuous fatigue functional. These hold for cyber-physical processes; transfer to discrete generative systems (LLM agents and similar) is not clean: token-level policies do not exhibit multiplicative weight decay, the embedding analogue of separation margin is not continuous along a trajectory, and the operational meaning of enrichment is undefined. Applicability to discrete generative systems is an open question.

STAK-PSAL does not prove inevitability of rupture. The monitor predicts bounded rupture horizon under continued degradation dynamics within the evaluated model family. Sufficient external enrichment, operator intervention, or structural modification may still restore admissibility before collapse.

Conclusion

A runtime system may remain operationally closed while already possessing a bounded rupture horizon. The machine-checked semantic hierarchy (`formal/metastable_recovery.v`) formalises this as the admissible visibility failure theorem: operational closure does not entail structural safety, and metastable closure admits constructive rupture witnesses bounded within a declared horizon. The machine-checked existential bridge (`formal/threshold_bridge.v`) connects the strict-equality formal layer to the ε-admissible operational layer, guaranteeing threshold crossing under geometric contraction; the explicit logarithmic ceiling bound of Lemma 1 is the quantitative refinement isolating the precise horizon. STAK-PSAL operationalises these results: $\mathcal{F}_n$ tracks cumulative structural degradation invisible to instantaneous monitors, with CRITICAL signals providing mean lead time 11.6 to 13.7 steps and zero false negatives in the 2,159 evaluated ruptured runs; and the three-witness certificate emits a bounded rupture witness on confirmed collapse. The Metastable Recovery Principle identifies the risk static and output-level monitors cannot detect: $\mathcal{F}_n$ is 13.8–14.9× higher in metastably closed runs. Runtime assurance is therefore path-dependent rather than purely state-dependent: admissibility may hold at time t while the history required to preserve it still carries a bounded rupture horizon. Safety-relevant distinguishability must therefore be monitored as a runtime quantity. Structural fatigue is not visible in the current output alone; metastable closure is detected only when the monitor retains memory of the path by which closure was reached.

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